

Homework 1**Q1.** What is the variable in the expression $2x + 3$?

- (A) 2
- (B) 3
- (C) x
- (D) $2x$
- (E) $2 + 3$

Q2. Identify the coefficient in the term $4y$

- (A) y
- (B) 4
- (C) $4y$
- (D) 1
- (E) 0

Q3. What is the constant term in the expression $2x - 5$?

- (A) $2x$
- (B) -5
- (C) 2
- (D) 5
- (E) $-2x$

Q4. In the expression $3 + 2$, what is the operator?

- (A) 3
- (B) 2
- (C) +
- (D) -
- (E) *

Q5. What does the term $2x^2$ represent?

- (A) The coefficient of x is 2 and the exponent is 2
- (B) The variable is 2 and the coefficient is x^2
- (C) The coefficient of x is 2 and x is squared
- (D) The coefficient is x and the variable is 2
- (E) The constant term is 2 and the variable term is x^2

Q6. In the expression $x + 2$, what operation is being performed?

- (A) Subtraction
- (B) Addition
- (C) Multiplication
- (D) Division
- (E) Exponentiation

Q7. Identify the term that is a variable in $3x + 2y - 4$

- (A) $3x$ and $2y$
- (B) $2y$ only
- (C) $3x$ only
- (D) -4
- (E) $3x + 2y$

Q8. What is the purpose of the order of operations?

- (A) To simplify fractions
- (B) To solve equations
- (C) To evaluate expressions with multiple operations
- (D) To graph functions
- (E) To find the perimeter of a shape

Open Ended Questions (FRQ)**Q9.** Draw and explain how you would represent the expression $2x + 1$ using algebra tiles. Be sure to include a key for your drawing.**Q10.** Explain, in your own words, what a variable is in an algebraic expression. Provide an example of a simple expression that contains a variable and describe what the variable represents in the context of the expression.**Chef's Tips**

- Focus on introducing basic concepts and definitions related to variables and expressions, such as coefficients and constants.
- Use simple, one-step problems to help students understand the fundamentals of the order of operations.
- Encourage students to use visual aids like algebra tiles to represent expressions and enhance their understanding of variables and expressions.